

SQL Cheat Sheet

Basic Commands

```
SHOW DATABASES;
USE database_name;
SHOW TABLES;
CREATE DATABASE db_name;
DROP DATABASE db_name;
```

Table Operations

```
CREATE TABLE Employees (...);
DESCRIBE Employees;
DROP TABLE Employees;
ALTER TABLE Employees RENAME TO Staff;
```

CRUD Operations

```
INSERT INTO Employees (...);
SELECT * FROM Employees;
UPDATE Employees SET Salary = 95000 WHERE EmployeeID = 1;
DELETE FROM Employees WHERE EmployeeID = 1;
```

Filtering Data

```
WHERE Salary > 80000 AND DepartmentID = 10;
LIKE 'A%'; -- starts with A
IN (10, 20, 30);
BETWEEN 60000 AND 100000;
```

Sorting & Limiting

```
ORDER BY Salary DESC;
LIMIT 5;
```

Aggregate Functions

```
COUNT(*), AVG(Salary), SUM(Salary), MIN(Salary), MAX(Salary)
```

GROUP BY & HAVING

```
GROUP BY DepartmentID;
HAVING COUNT(*) > 5;
```

Joins

```
INNER JOIN Departments d ON e.DepartmentID = d.DepartmentID;
LEFT JOIN Departments d ON e.DepartmentID = d.DepartmentID;
RIGHT JOIN Departments d ON e.DepartmentID = d.DepartmentID;
```

CASE (Conditional Logic)

```
CASE WHEN Salary >= 80000 THEN 'High' WHEN Salary >= 60000 THEN 'Medium' ELSE 'Low'  
END AS SalaryLevel
```

Subqueries

```
WHERE Salary > (SELECT AVG(Salary) FROM Employees);
```

Constraints

```
PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL
```